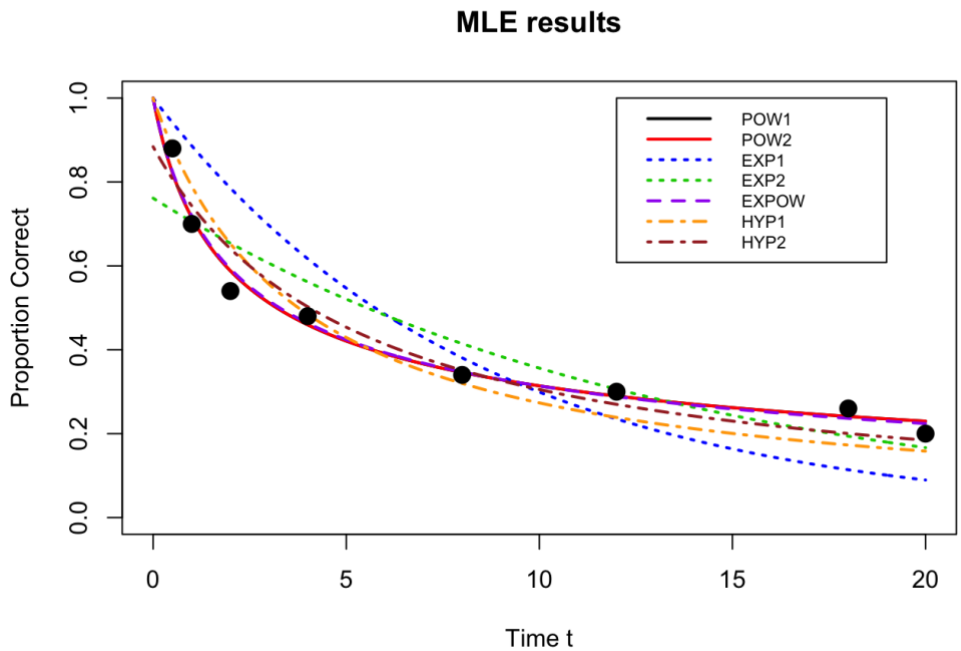
HW1_Dongwook Hwang

MLE

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[1] "- MLE results ------------"
  Models   loglik    r2
1   POW1 -235.406 0.980
2   POW2 -235.406 0.980
3   EXP1 -258.926 0.596
4   EXP2 -241.001 0.865
5  EXPOW -235.389 0.980
6   HYP1 -238.774 0.911
7   HYP2 -237.264 0.942
```

```
[1] "- Best-fit parameters --------"
      POW1  POW2  EXP1  EXP2 EXPOW  HYP1  HYP2
par1 0.483 1.000 0.121 0.761 1.000 0.266 0.884
par2    NA 0.483    NA 0.076 0.003    NA 0.190
par3    NA    NA    NA    NA 0.469    NA    NA
```

The MLE results show that the power-law models (POW1 and POW2) and the hybrid EXPOW model provide the best fit to the retention data, all achieving $R^2 = 0.980$. Notably, POW2's estimated parameter a = 1.000 indicates that it effectively reduces to the simpler POW1 model (since a × (t+1)^(-β) becomes (t+1)^(-β) when a = 1). Similarly, EXPOW's estimated $\beta = 0.003$ is nearly zero, meaning its exponential component contributes almost nothing, and the model is essentially driven by the power-law component alone ($\gamma = 0.469$, close to POW1's $\beta = 0.483$). This suggests that the power-law decay function best captures the forgetting pattern in this dataset. Among the remaining models, HYP2 ($R^2 = 0.942$) and HYP1 ($R^2 = 0.911$) show moderate fits, while EXP2 ($R^2 = 0.865$) performs poorly and EXP1 ($R^2 = 0.596$) provides the worst fit, indicating

that a purely exponential decay is insufficient to describe the observed retention curve. As a result, forgetting follows a power-law rather than an exponential function.

LSE

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[1] "- LSE results ------------"
  Models   SSE    r2
1   POW1 0.008 0.980
2   POW2 0.008 0.980
3   EXP1 0.151 0.611
4   EXP2 0.052 0.867
5  EXPOW 0.008 0.980
6   HYP1 0.034 0.912
7   HYP2 0.022 0.943
[1] "- LSE Best-fit parameters --------"
      POW1  POW2  EXP1 EXP2 EXPOW  HYP1  HYP2
par1 0.486 1.000 0.139 0.76 1.000 0.276 0.872
par2    NA 0.486    NA 0.08 0.001    NA 0.190
par3    NA    NA    NA   NA 0.483    NA    NA
```

Additionally, Least Squares Estimation (LSE) was also conducted for all seven models. The LSE results were highly consistent with the MLE results: the model ranking remained identical, with POW1, POW2, and EXPOW achieving the highest R^2 (0.980) and EXP1 showing the worst fit (R^2 = 0.611). The estimated parameters were similar but not identical across the two methods — for example, POW1's β was 0.483 (MLE) vs. 0.486 (LSE). This difference arises because, unlike MLE, LSE simply minimizes the squared distance between observed and predicted proportions. The convergence of both methods on the same model ranking strengthens the conclusion that power-law models provide the most appropriate description of this retention data.